

Real-World Order of Operations

MA.8.NSO.1.7

I can substitute values into a formula and evaluate it, reading what the situation does rather than hunting for key words.

Yesterday you read an expression's structure.

Today the expression arrives wrapped in a sentence — and the sentence is not a list of key words.

Warm-Up

On your own. Four minutes. No notes.

1. $-5^2 + (-5)^2$
2. $\sqrt{144} + 2^{-1}$
3. Find the value: $\left(\frac{3}{4}\right)^{-1}$
4. Evaluate $2x^2$ when $x = 3$.

Warm-Up

Answers.

1. $-5^2 + (-5)^2$ **0**

2. $\sqrt{144} + 2^{-1}$ **25/2**

3. Find the value: $\left(\frac{3}{4}\right)^{-1}$ **4/3**

4. Evaluate $2x^2$ when $x = 3$. **18**

Notes

Copy both lines.

I. Substituting: the value goes in parentheses

$$K = \frac{1}{2}mv^2 \quad \text{with} \quad m = 60 \quad \text{and} \quad v = 10:$$

$$\frac{1}{2}(60)(10)^2 = \frac{1}{2}(60)(100) = 3,000 \quad \text{joules}$$

$$\text{with } x = -4: \quad x^2 = (-4)^2 = 16, \quad \text{never} \quad -4^2 = -16$$

Substitute in parentheses, then evaluate by the structure — powers before products.

Notes

Copy both problems.

II. Key words do not decide the operation

15 dollars plus 4 dollars for each of 6 gigabytes:

$$15 + 4(6) = 39$$

15 dollars for each of 4 lines, plus 6 dollars of tax:

$$15(4) + 6 = 66$$

The same words, the same numbers, different structures. Read the situation.

Notes

III. Round once, at the end — and say the unit

$$6\sqrt{200} = 84.8528... \approx 84.85 \text{ feet}$$

rounding the side to 14 first: $6(14) = 84 \text{ feet}$

Carry the exact value, round once at the end, and write the unit.

Example 1

A formula with a power in it.

Kinetic energy is $K = \frac{1}{2}mv^2$,

for mass m in kilograms and speed v in metres per second.

Find K for a car of mass 1,200 kg travelling at 15 m/s.

Give the answer with its unit.

Example 1

Worked.

$$K = \frac{1}{2}(1,200)(15)^2 \quad \text{substitute, each value in parentheses}$$

$$= \frac{1}{2}(1,200)(225) \quad \text{the power first — it belongs to the 15 alone}$$

$$= 600(225) = 135,000 \quad \text{then the products}$$

Answer: 135,000 joules

Your Turn

Same steps, your numbers. Boards up when done.

Find $K = \frac{1}{2}mv^2$ for $m = 850$ kg and $v = 12$ m/s.

Your Turn

Answer.

Find $K = \frac{1}{2}mv^2$ for $m = 850$ kg and $v = 12$ m/s.

425 times 144

Answer: 61,200 joules

Example 2

A cost with a rate and a flat fee.

A square floor of side 9 feet is carpeted.

Carpet is 2.89 dollars a square foot and padding is 0.75 dollars a square foot.

Installation is a flat 95 dollars. Find the total cost.

Example 2

Worked.

$$9^2(2.89 + 0.75) + 95 \quad \text{area times the per-square-foot prices, then the flat fee}$$

$$= 81(3.64) + 95 \quad \text{the area and the rate, each finished}$$

$$= 294.84 + 95 = 389.84 \quad \text{multiply, then add}$$

Answer: 389.84 dollars

Your Turn

Same steps, your numbers. Boards up when done.

A square bathroom floor of side 6 feet is tiled.

Tile is 4.25 dollars a square foot and grout is a flat 13.48 dollars.

Find the total cost.

Your Turn

Answer.

A square bathroom floor of side 6 feet is tiled.

Tile is 4.25 dollars a square foot and grout is a flat 13.48 dollars.

Find the total cost.

$36(4.25)$ then add

Answer: 166.48 dollars

Whiteboards • Question 1 of 9

Boards up on three.

Evaluate $3x^2$ when $x = -2$.

Answer it.

Substitute in parentheses.

Whiteboards • Question 1 of 9

Answer.

Evaluate $3x^2$ when $x = -2$.

$$3(-2)^2$$

Answer: 12

Whiteboards • Question 2 of 9

Boards up on three.

A cyclist of mass $m = 70$ kg rides at $v = 8$ m/s.

Her bicycle cost 480 dollars.

Find her kinetic energy, $K = \frac{1}{2}mv^2$.

Answer it.

Substitute in parentheses.

Whiteboards • Question 2 of 9

Answer.

A cyclist of mass $m = 70$ kg rides at $v = 8$ m/s.

Her bicycle cost 480 dollars.

Find her kinetic energy, $K = \frac{1}{2}mv^2$.

35 times 64

Answer: 2,240 joules

Whiteboards • Question 3 of 9

Boards up on three.

A plan costs 15 dollars plus 4 dollars for each gigabyte.
Find the cost of 6 gigabytes.

Answer it.

What happens six times?

Whiteboards • Question 3 of 9

Answer.

A plan costs 15 dollars plus 4 dollars for each gigabyte.

Find the cost of 6 gigabytes.

$$15 + 4(6)$$

Answer: 39 dollars

Whiteboards • Question 4 of 9

Boards up on three.

A square floor of side 8 feet is tiled at 3.50 dollars a square foot.

Installation is a flat 60 dollars.

Find the total cost.

Answer it.

Which price depends on the area?

Whiteboards • Question 4 of 9

Answer.

A square floor of side 8 feet is tiled at 3.50 dollars a square foot.

Installation is a flat 60 dollars.

Find the total cost.

$$64(3.50) + 60$$

Answer: 284 dollars

Whiteboards • Question 5 of 9

Boards up on three.

Evaluate $\frac{1}{2}mv^2$ when $m = 40$ and $v = 3$.

Which is the value?

A. 180

B. 120

C. 360

D. 3,600

Answer it.

Whiteboards • Question 5 of 9

Answer.

Evaluate $\frac{1}{2}mv^2$ when $m = 40$ and $v = 3$.

Which is the value?

A. 180

B. 120

C. 360

D. 3,600

Answer: A — 180

Whiteboards • Question 6 of 9

Boards up on three.

A square patio of side 12 feet is sealed at 1.25 dollars a square foot.

Delivery of the sealer is a flat 38 dollars.

Which expression gives the total cost?

A. $12(1.25) + 38$

B. $12^2 + 1.25 + 38$

C. $12^2(1.25 + 38)$

D. $12^2(1.25) + 38$

Answer it.

Whiteboards • Question 6 of 9

Answer.

A square patio of side 12 feet is sealed at 1.25 dollars a square foot.

Delivery of the sealer is a flat 38 dollars.

Which expression gives the total cost?

A. $12(1.25) + 38$

B. $12^2 + 1.25 + 38$

C. $12^2(1.25 + 38)$

D. $12^2(1.25) + 38$

Answer: D — $12^2(1.25) + 38$

Whiteboards • Question 7 of 9

Boards up on three.

A fence runs along 6 sides, each $\sqrt{200}$ feet long.

Find its total length, to the nearest hundredth of a foot.

Answer it.

Round once, at the end.

Whiteboards • Question 7 of 9

Answer.

A fence runs along 6 sides, each $\sqrt{200}$ feet long.

Find its total length, to the nearest hundredth of a foot.

$$6 \times 14.1421\dots$$

Answer: 84.85 feet

Whiteboards • Question 8 of 9

Boards up on three.

Evaluate $\frac{a^2 - b}{c}$ when $a = 7$, $b = 4$, $c = 5$.

Answer it.

Finish the top first.

Whiteboards • Question 8 of 9

Answer.

Evaluate $\frac{a^2 - b}{c}$ when $a = 7$, $b = 4$, $c = 5$.

45 over 5

Answer: 9

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

A plan costs 15 dollars plus 4 dollars for each of 6 gigabytes.

Another costs 15 dollars for each of 4 lines, plus 6 dollars of tax.

Find both totals, and say why the key words could not have told you.

Write your answer in sentences.

The same three numbers, the same words.

Whiteboards • Question 9 of 9

Answer.

A plan costs 15 dollars plus 4 dollars for each of 6 gigabytes.

Another costs 15 dollars for each of 4 lines, plus 6 dollars of tax.

Find both totals, and say why the key words could not have told you.

39 and 66

Answer: 39 dollars and 66 dollars

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate variable expressions: word problems — DPG